Homework 2

Autonomous Mobile Robots

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Instructions

This document provides the results of the codes of the assignment provided in the Codes folder. The file names are appropriately labelled to indicate the question they are presenting the solution for. For questions 5 and 6 there are 2 different codes displaying the simulation in both 2D and 3D plots. The solutions for the extra credit questions are in the folder named extra. Please note that all the different constants used in the codes are chosen to produce optimal results and they may not be consistent across codes. However I believe that the program logic is correct irrespective of the constants chosen. Different constants can be chosen to produce different results and behaviours. The velocity for the robots is updated by finding the negative derivative of the potential field which gives us the force in x and y which can then be used as acceleration assuming the mass is 1. In the radial potential field scenario I had to use a very large repulsive potential constant value to actually make the point mass's path to be out of the square obstacle.

1

Create a 100mx100m workspace

Solution

Solution utilized in other questions.

2

Create an attractive potential filed centered at q_{goal}=(80,20); Plot the potential field

Solution

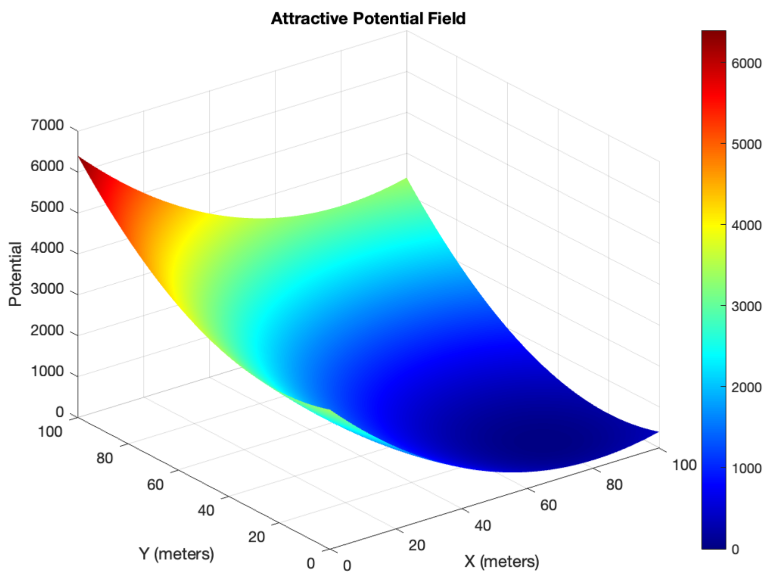

Figure 1: Attractive Potential Field

3

Create a square obstacle with the following coordinates (30,50)(50,50)(50,70)(30,70) and implement a radial repulsive potential field centered in the obstacle; Plot the potential field.

Solution

In this question the field is assigned a maximum values so as to prevent the field from reaching infinity and being too narrow and reducing the impact of attractive field in the combined graph. That is the reason why the field looks cut off at the top.

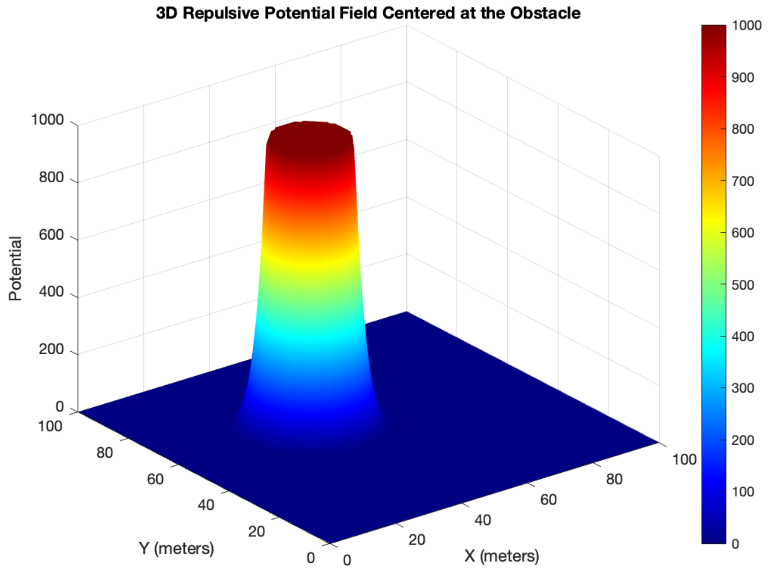

Figure 2: Radial Repulsive Potential Field

4

Combine the two potential fields and plot the resulting field

Solution

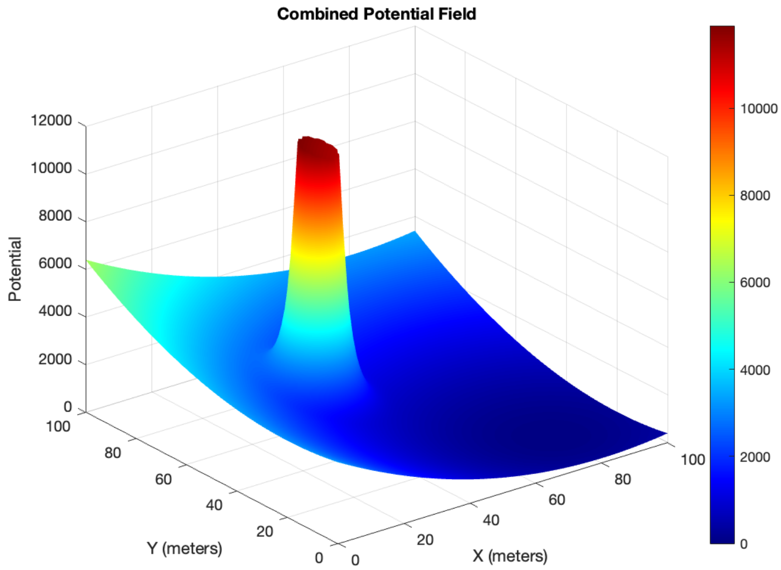

Figure 3: Combined Potential Field

5

Create a simple point-mass holonomic robot centered at q_{start}=(10,80) and use the resulting field to navigate the robot from q_{start} to q_{goal}. V_{max} = 3m/s. Plot the vehicle trajectory in the workspace and show the animation as the robot navigates to the goal.

Solution

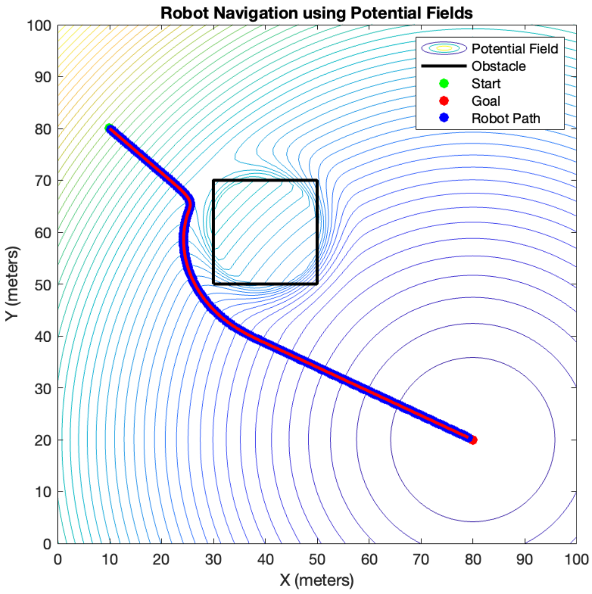

Figure 4: Holonomic Point Mass Navigation To Goal 2D

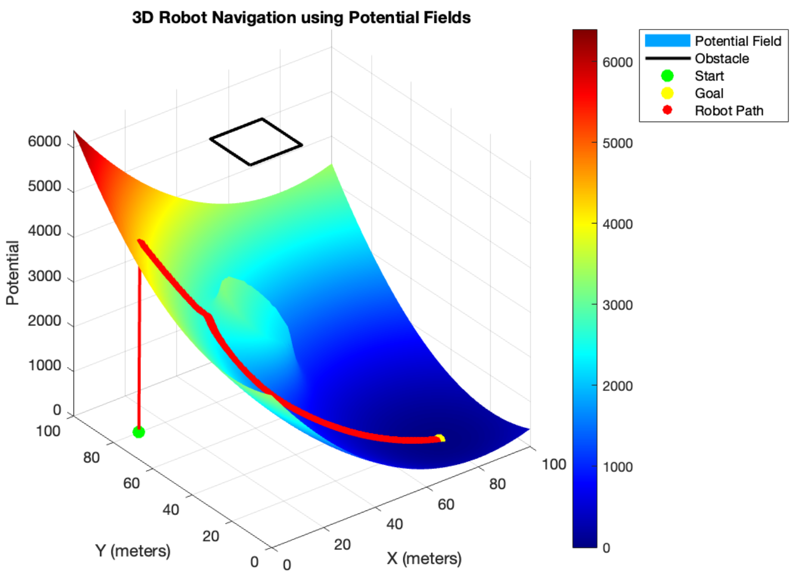

Figure 5: Holonomic Point Mass Navigation To Goal 3D

6

Repeat 5. using the non-holonomic robot designed in HW-1

Solution

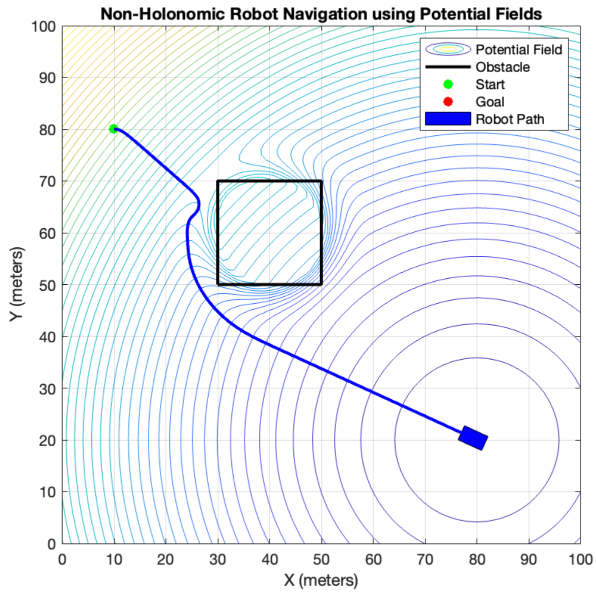

Figure 6: Non-Holonomic Robot Navigation To Goal 2D

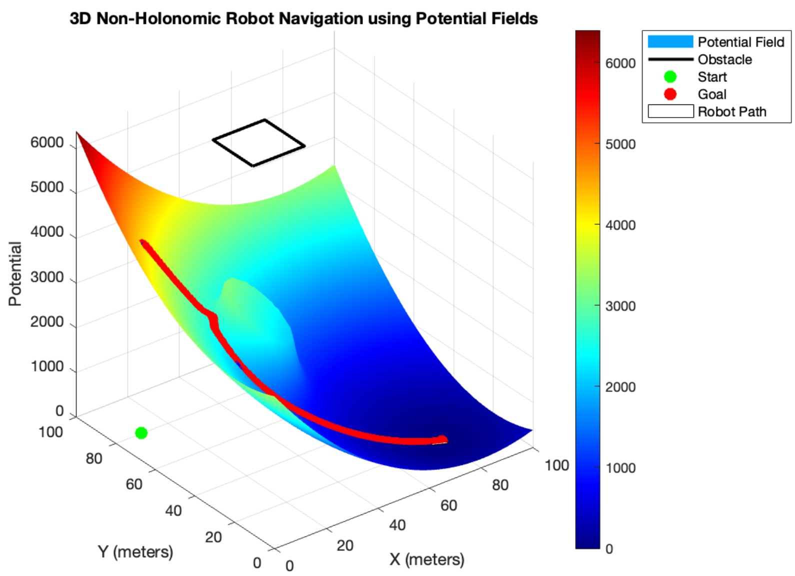

Figure 7: Non-Holonomic Robot Navigation To Goal 3D

7

Repeat 3-7 implementing a repulsive potential filed that follows the shape of the obstacle

Solution

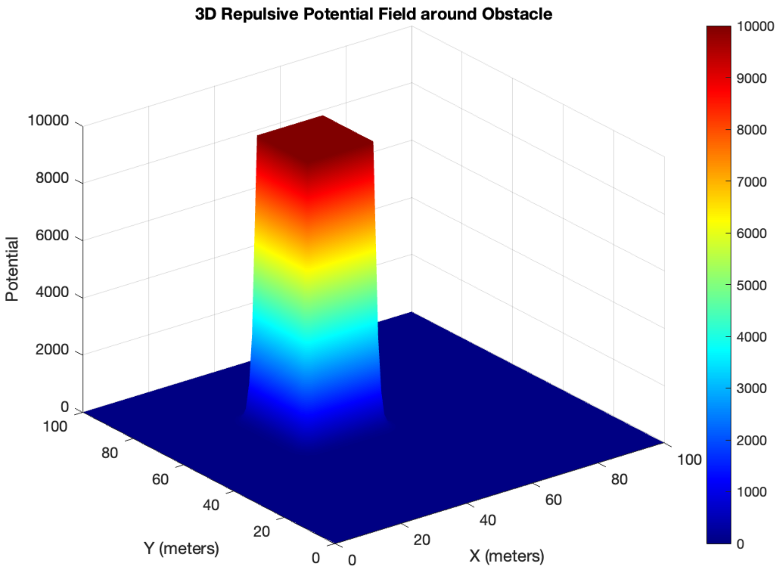

Figure 8: Repulsive Potential Field Around The Obstacle

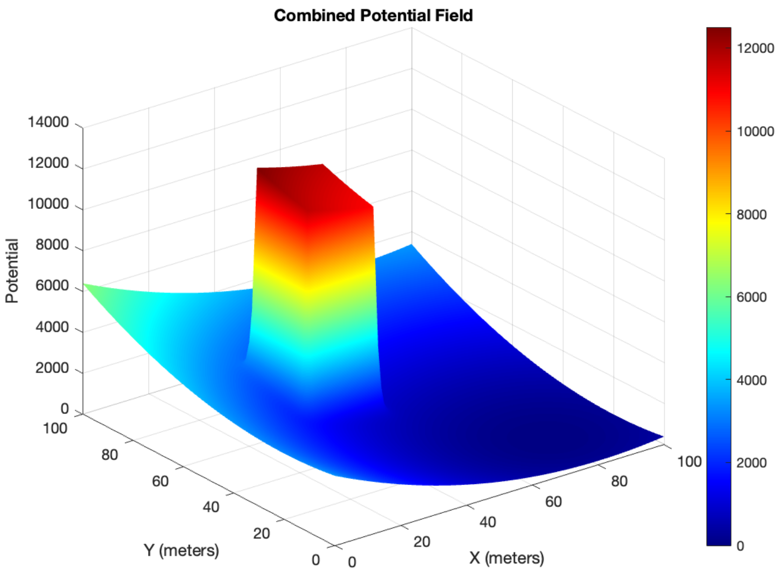

Figure 9: Combined Potential Field

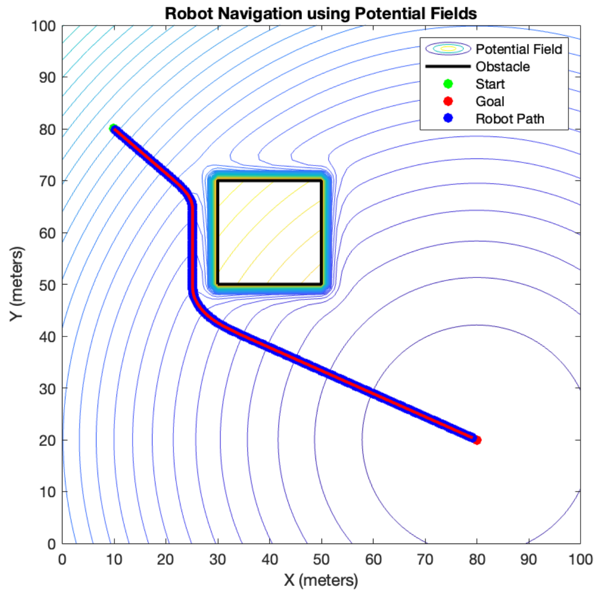

Figure 10: Holonomic Point Mass Navigation To Goal 2D

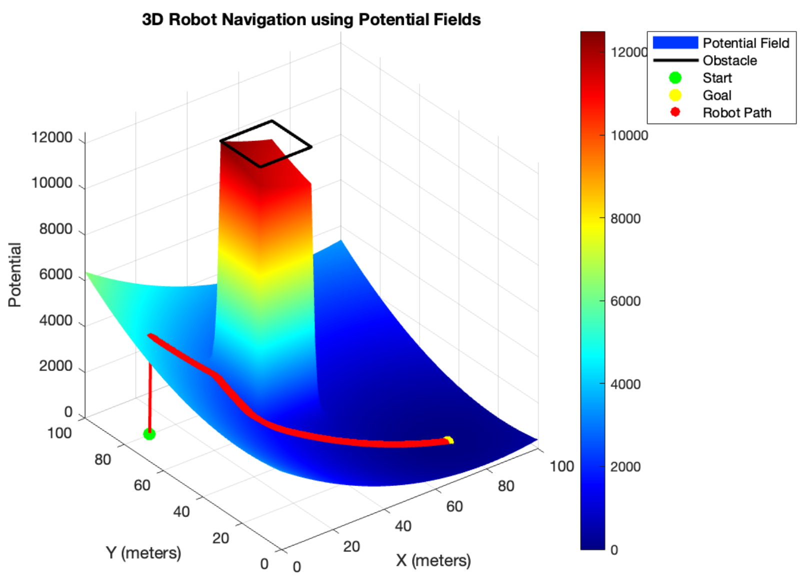

Figure 11: Holonomic Point Mass Navigation To Goal 3D

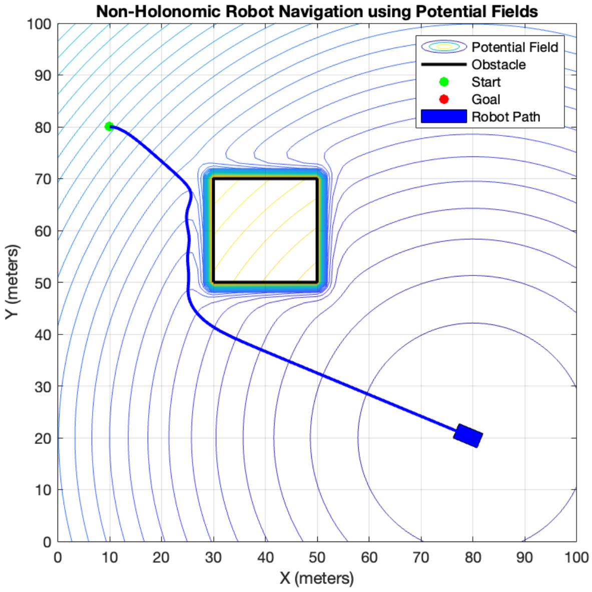

Figure 12: Non-Holonomic Robot Navigation To Goal 2D

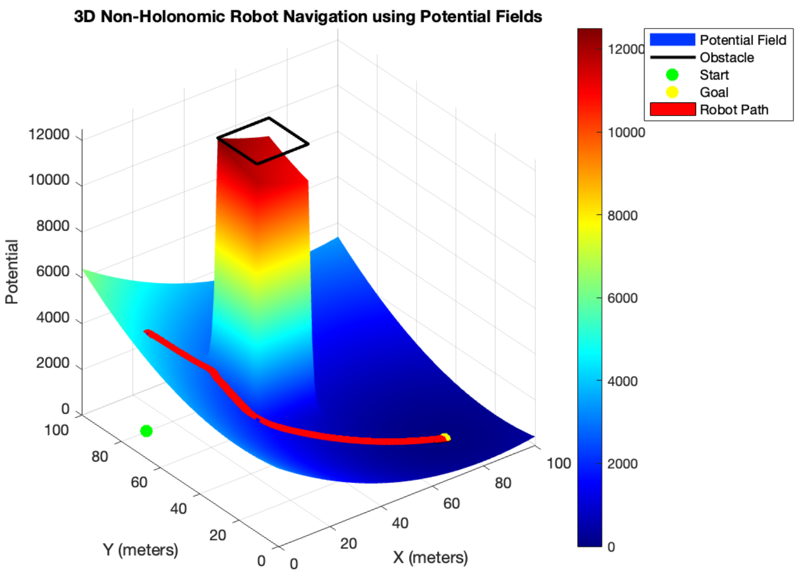

Figure 13: Non-Holonomic Robot Navigation To Goal 3D